

SUPPLEMENTARY MATERIAL FOR “CAUSAL DISCOVERY FROM COMPLEX SURVEY DATA: A SURVEY-WEIGHTED ORDINAL BAYESIAN NETWORK FOR SOCIAL DETERMINANTS OF HEALTH”

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This supplement provides: (A) full proofs of all theoretical results stated in the main text; (B) details of the polychoric clustering procedure; (C) simulation Study 3 results for survey-weighted PC, FCI, and LiNGAM; (D) the complete bootstrap edge probability matrix from the application; and (E) survey weight distribution details. References to assumptions, propositions, and theorems refer to the main text unless otherwise noted.

1. The swa-oBn algorithm.

Algorithm 1 swa-oBn: Survey-weighted, covariate-adjusted ordinal causal discovery

Require: Ordinal data $\mathbf{X} \in \mathbb{R}^{n \times q}$; survey weights $\mathbf{w} = (w_1, \dots, w_n)$; covariates $\mathbf{Z} \in \mathbb{R}^{n \times r}$ (optional); blacklist $\mathcal{B}$, whitelist $\mathcal{W}$ (optional); number of restarts R ; number of bootstrap replicates B

Ensure: Point-estimate DAG $\hat{\mathcal{G}}$; edge inclusion probabilities $\{\hat{p}_{jk}\}$

{Preprocessing}

- 1: Normalize weights: $w_i \leftarrow w_i \cdot n / \sum_{i'} w_{i'}$
- 2: Compute $n_{\text{eff}} \leftarrow (\sum_i w_i)^2 / \sum_i w_i^2$
{Point estimate via multi-start hill-climbing}
- 3: **for** $s = 1, \dots, R$ **do**
- 4: Initialize $\mathcal{G}^{(s)}$ (empty graph if $s = 1$; random sparse DAG otherwise)
- 5: Enforce whitelist edges in $\mathcal{G}^{(s)}$
- 6: **repeat**
- 7: **Step 1 (Skeleton):** For each candidate add/delete move consistent with $\mathcal{B}$, $\mathcal{W}$, and acyclicity, evaluate the change in pBIC (Equation 7) using the covariate-adjusted weighted score (Equation 10)
- 8: **Step 2 (Orientation):** For each candidate reversal move consistent with $\mathcal{B}$, $\mathcal{W}$, and acyclicity, evaluate the change in the unpenalized rescaled pseudo-log-likelihood
- 9: Apply the move with the largest improvement, if positive
- 10: **until** no improving move exists
- 11: **end for**
- 12: $\hat{\mathcal{G}} \leftarrow \arg \min_s \text{pBIC}(\mathcal{G}^{(s)})$
{Bootstrap uncertainty quantification}
- 13: **for** $b = 1, \dots, B$ **do**
- 14: Draw indices $\mathcal{I}^{(b)}$ by sampling $\{1, \dots, n\}$ with replacement
- 15: Construct $(\mathbf{X}^{(b)}, \mathbf{w}^{(b)}, \mathbf{Z}^{(b)})$ from $\mathcal{I}^{(b)}$; recompute $n_{\text{eff}}^{(b)}$
- 16: Run lines 3–12 on $(\mathbf{X}^{(b)}, \mathbf{w}^{(b)}, \mathbf{Z}^{(b)})$ to obtain $\hat{\mathcal{G}}^{(b)}$
- 17: **end for**
- 18: $\hat{p}_{jk} \leftarrow B^{-1} \sum_{b=1}^B \mathbf{1}(j \rightarrow k \in \hat{\mathcal{G}}^{(b)})$ for all (j, k)

2. Proofs of Theoretical Results. Throughout, we use the notation established in Section 5 of the main text: ξ denotes the superpopulation, U_N the finite population of size N , S_n the probability sample of size n , $w_i = 1/\pi_i$ the survey weights, and $P_\xi(\mathbf{X} \mid \mathcal{G}, \theta, \mathbf{Z})$ the ordinal BN model under DAG $\mathcal{G}$ with parameters θ and covariates $\mathbf{Z}$. We define:

- Census log-likelihood: $\ell_U(\theta \mid \mathcal{G}) = \sum_{i \in U_N} \log P(\mathbf{x}_i \mid \mathcal{G}, \theta, \mathbf{z}_i)$.
- Pseudo-log-likelihood: $\tilde{\ell}(\theta \mid \mathcal{G}) = \sum_{i \in S_n} w_i \log P(\mathbf{x}_i \mid \mathcal{G}, \theta, \mathbf{z}_i)$.
- Superpopulation expected log-likelihood: $Q_\xi(\theta \mid \mathcal{G}) = \mathbb{E}_\xi[\log P(\mathbf{X} \mid \mathcal{G}, \theta, \mathbf{Z})]$.
- Normalized versions: $\bar{\ell}_U(\theta \mid \mathcal{G}) = N^{-1} \ell_U(\theta \mid \mathcal{G})$, $\bar{\tilde{\ell}}(\theta \mid \mathcal{G}) = N^{-1} \tilde{\ell}(\theta \mid \mathcal{G})$.

We use $\theta_{\mathcal{G}}^*$ to denote the superpopulation maximizer of $Q_\xi(\cdot \mid \mathcal{G})$, i.e., $\theta_{\mathcal{G}}^* = \arg \max_{\theta} Q_\xi(\theta \mid \mathcal{G})$; when $\mathcal{G} = \mathcal{G}^*$, this coincides with the true data-generating parameter, which we denote simply by θ^* . For a misspecified DAG $\mathcal{G}' \neq \mathcal{G}^*$, we use $\theta_{\mathcal{G}'}^\dagger \equiv \theta_{\mathcal{G}'}^*$ to emphasize its role as the KL-projection rather than a truth-level parameter. The *conditional* (covariate-pointwise) KL-projection is denoted $\theta^\dagger(\mathbf{z}) = \arg \max_{\theta} \sum_{\mathbf{x}} P(\mathbf{x} \mid \mathcal{G}^*, \theta^*, \mathbf{z}) \log P(\mathbf{x} \mid \mathcal{G}', \theta, \mathbf{z})$, which in general differs from the unconditional $\theta_{\mathcal{G}'}^\dagger$.

2.1. Proof of Proposition 1 (Identifiability under Covariate Adjustment). Let $\mathcal{G}^* \neq \mathcal{G}'$ be Markov-equivalent DAGs. Write $\vartheta = (\theta, \delta) \in \Theta \times \mathbb{R}^{r_q}$ for the full parameter under the covariate-adjusted model, and define the pointwise pseudo-true parameter

$$\theta^\dagger(\mathbf{z}; \vartheta^*) = \arg \max_{\theta} \sum_{\mathbf{x}} P(\mathbf{x} \mid \mathcal{G}^*, \vartheta^*, \mathbf{z}) \log P(\mathbf{x} \mid \mathcal{G}', \theta, \mathbf{z}).$$

Define the exceptional set

$$\mathcal{E} = \left\{ \vartheta^* : \mathbb{E}_{\mathbf{Z}} \left[\text{KL} \left(P_\xi(\mathbf{X} \mid \mathcal{G}^*, \vartheta^*, \mathbf{Z}) \parallel P(\mathbf{X} \mid \mathcal{G}', \theta^\dagger(\mathbf{Z}; \vartheta^*), \mathbf{Z}) \right) \right] = 0 \right\}.$$

We show that $\mathcal{E}$ has Lebesgue measure zero in $\Theta \times \mathbb{R}^{r_q}$.

2.1.1. Step 1: Reduction to the no-covariate case.. For each fixed $\mathbf{Z} = \mathbf{z}$, the covariate-adjusted cumulative link model becomes

$$P_\xi(X_j \leq \ell \mid \mathbf{X}_{\text{pa}(j)}, \mathbf{Z} = \mathbf{z}) = F \left(\gamma_{j\ell} - \sum_{k \in \text{pa}(j)} \beta_{jk} X_k - \tilde{\alpha}_j(\mathbf{z}) \right),$$

with $\tilde{\alpha}_j(\mathbf{z}) = \alpha_j + \delta_j' \mathbf{z}$. This is a no-covariate oBN with thresholds $\{\gamma_{j\ell}\}$, parent-effect parameters $\{\beta_{jk}\}$, and shifted intercepts $\{\tilde{\alpha}_j(\mathbf{z})\}$.

2.1.2. Step 2: The identifiability variety is intercept-invariant.. The identifiability proof of Ni and Mallick (2022) establishes that the ordinal parameterizations $P(X_j \mid X_k)$ and $P(X_k \mid X_j)$ are generically non-equivalent by analyzing a polynomial system in the threshold and parent-effect parameters. The intercept α_j determines the marginal location of the conditional distribution but does not enter the polynomial relations that govern distributional equivalence. Consequently, the exceptional set

$$\mathcal{V} = \left\{ (\gamma, \beta, \alpha) : P(\mathbf{X} \mid \mathcal{G}^*, \gamma, \beta, \alpha) \equiv P(\mathbf{X} \mid \mathcal{G}', \theta_{\text{NC}}^\dagger(\gamma, \beta, \alpha)) \right\}$$

in the no-covariate parameter space is a proper algebraic subvariety of Lebesgue measure zero, invariant under intercept translation. Here $\theta_{\text{NC}}^\dagger$ denotes the no-covariate pseudo-true parameter. By Assumption 5, $\mathcal{V}$ has Lebesgue measure zero.

2.1.3. *Step 3: The exceptional set has measure zero in (θ^*, δ^*) -space..* Case (a): $\mathbf{Z}$ is continuous with a density on $\mathbb{R}^r$. Fix (θ^*, δ^*) and consider the map $\mathbf{z} \mapsto (\gamma^*, \beta^*, \tilde{\alpha}^*(\mathbf{z}))$. This is an affine map. Since $\mathcal{V}$ has Lebesgue measure zero and the affine map has nonzero Jacobian for δ^* outside a measure-zero degenerate set, the pre-image $\{\mathbf{z} : (\gamma^*, \beta^*, \tilde{\alpha}^*(\mathbf{z})) \in \mathcal{V}\}$ has Lebesgue measure zero in $\mathbf{z}$ -space. Therefore the pointwise KL is strictly positive for $P_{\mathbf{Z}}$ -almost every $\mathbf{z}$, and

$$\mathbb{E}_{\mathbf{Z}} \left[\text{KL} \left(P_{\xi}(\cdot \mid \mathcal{G}^*, \vartheta^*, \mathbf{Z}) \parallel P(\cdot \mid \mathcal{G}', \theta^\dagger(\mathbf{Z}; \vartheta^*), \mathbf{Z}) \right) \right] > 0$$

whenever δ^* avoids this degenerate set. Case (b): $\mathbf{Z}$ is discrete with finite support $\{\mathbf{z}_1, \dots, \mathbf{z}_K\}$. For each k , define

$$\mathcal{E}_k = \left\{ (\theta^*, \delta^*) : (\gamma^*, \beta^*, \alpha_j^* + (\delta_j^*)^\top \mathbf{z}_k) \in \mathcal{V} \right\}.$$

For each fixed $\mathbf{z}_k$, the map $(\theta^*, \delta^*) \mapsto (\gamma^*, \beta^*, \alpha_j^* + (\delta_j^*)^\top \mathbf{z}_k)$ is affine with nonzero Jacobian (provided $\mathbf{z}_k \neq \mathbf{0}$). By Fubini's theorem and the measure-zero property of $\mathcal{V}$, $\mathcal{E}_k$ has Lebesgue measure zero in (θ^*, δ^*) -space. The overall exceptional set satisfies $\mathcal{E} \subseteq \bigcup_{k=1}^K \mathcal{E}_k$, which, as a finite union of measure-zero sets, has Lebesgue measure zero. (When $\mathbf{z}_k = \mathbf{0}$ for some k , we redefine the origin so that all $\mathbf{z}_k$ are nonzero; this does not change the model.) Therefore $\mathcal{E}$ has Lebesgue measure zero, as claimed. When the true parameter $(\theta^*, \delta^*) \notin \mathcal{E}$, the expected KL divergence is strictly positive. $\square$

2.2. *Proof of Proposition 2 (Pseudo-MLE Consistency).* We prove that $\hat{\theta}_{\mathcal{G}}^w \xrightarrow{p} \theta_{\mathcal{G}}^*$ under the joint $(\xi \times p_N)$ -probability. The argument proceeds in three steps.

2.2.1. *Step 1: Design consistency of the normalized pseudo-log-likelihood..* Fix $\theta \in \Theta_{\mathcal{G}}$ and define $g_{\theta}(\mathbf{x}, \mathbf{z}) = \log P(\mathbf{x} \mid \mathcal{G}, \theta, \mathbf{z})$. Since $\mathbf{X}$ takes values in the finite set $\prod_{j=1}^q \{1, \dots, L_j\}$, Assumption 6 (Non-Degeneracy) gives $P(\mathbf{x} \mid \mathcal{G}, \theta, \mathbf{z}) > 0$ for all $(\mathbf{x}, \mathbf{z})$, and compactness of $\Theta_{\mathcal{G}}$ (Assumption 10(i)) combined with continuity of F yields a uniform lower bound $P(\mathbf{x} \mid \mathcal{G}, \theta, \mathbf{z}) \geq p_{\min} > 0$ over all $(\mathbf{x}, \mathbf{z}, \theta)$. Therefore g_{θ} is bounded uniformly in $\theta \in \Theta_{\mathcal{G}}$. By Assumption 9 (Design Consistency), applied to the function g_{θ} of $(\mathbf{x}, \mathbf{z})$,

$$(1) \quad \bar{\ell}(\theta \mid \mathcal{G}) - \bar{\ell}_U(\theta \mid \mathcal{G}) = \frac{1}{N} \sum_{i \in S_n} w_i g_{\theta}(\mathbf{x}_i, \mathbf{z}_i) - \frac{1}{N} \sum_{i \in U_N} g_{\theta}(\mathbf{x}_i, \mathbf{z}_i) \xrightarrow{p_N} 0.$$

Design Consistency in turn follows from Horvitz–Thompson consistency under Assumptions 7–8 (Positivity, Bounded Weights); see Fuller (2009), Ch. 1.

2.2.2. *Step 2: Superpopulation law of large numbers for the census log-likelihood..* By the i.i.d. superpopulation structure, $\{(\mathbf{x}_i, \mathbf{z}_i)\}_{i \in U_N}$ are i.i.d. draws from ξ . Since g_{θ} is bounded, the strong law of large numbers gives

$$(2) \quad \bar{\ell}_U(\theta \mid \mathcal{G}) = \frac{1}{N} \sum_{i=1}^N g_{\theta}(\mathbf{x}_i, \mathbf{z}_i) \xrightarrow{\text{a.s.}} \mathbb{E}_{\xi}[g_{\theta}(\mathbf{X}, \mathbf{Z})] = Q_{\xi}(\theta \mid \mathcal{G}).$$

2.2.3. *Step 3: Uniform convergence and argmax convergence..* Combining (1) and (2), for each fixed θ : $\bar{\ell}(\theta \mid \mathcal{G}) \xrightarrow{p} Q_{\xi}(\theta \mid \mathcal{G})$. We upgrade this to uniform convergence over $\Theta_{\mathcal{G}}$. By Assumption 10(iv), for any $\theta, \theta' \in \Theta_{\mathcal{G}}$,

$$|g_{\theta}(\mathbf{x}, \mathbf{z}) - g_{\theta'}(\mathbf{x}, \mathbf{z})| \leq M \|\theta - \theta'\|$$

for some constant $M > 0$. This Lipschitz condition, combined with the compactness of $\Theta_{\mathcal{G}}$ (Assumption 10(i)), gives equicontinuity. Pointwise convergence plus equicontinuity on a compact set gives uniform convergence (van der Vaart, 1998, Theorem 2.4):

$$(3) \quad \sup_{\theta \in \Theta_{\mathcal{G}}} \left| \bar{\ell}(\theta | \mathcal{G}) - Q_{\xi}(\theta | \mathcal{G}) \right| \xrightarrow{p} 0.$$

By Assumption 10(iii), $Q_{\xi}(\theta | \mathcal{G})$ has a unique maximizer $\theta_{\mathcal{G}}^*$ with nonsingular Hessian. By (3) and the argmax continuous mapping theorem (van der Vaart, 1998, Theorem 5.7),

$$(4) \quad \hat{\theta}_{\mathcal{G}}^w \xrightarrow{p} \theta_{\mathcal{G}}^*.$$

As a corollary:

$$(5) \quad \bar{\ell}(\hat{\theta}_{\mathcal{G}}^w | \mathcal{G}) \xrightarrow{p} Q_{\xi}(\theta_{\mathcal{G}}^* | \mathcal{G}).$$

□

2.3. Proof of Theorem 1 (Skeleton Consistency). Let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ a DAG not Markov-equivalent to $\mathcal{G}^*$. We show $\Pr(\widehat{\text{BIC}}(\mathcal{G}^*) < \widehat{\text{BIC}}(\mathcal{G}')) \rightarrow 1$.

Write

$$(6) \quad \widehat{\text{BIC}}(\mathcal{G}^*) - \widehat{\text{BIC}}(\mathcal{G}') = -2 \left[\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) - \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}') \right] + (K_{\mathcal{G}^*} - K_{\mathcal{G}'})\rho_n.$$

2.3.1. Case 1A: $\mathcal{G}'$ is an overfit model ($\mathcal{G}^* \subset \mathcal{G}'$ as directed graphs, so $K_{\mathcal{G}'} > K_{\mathcal{G}^*}$). Since $\mathcal{G}^*$ is a subgraph of $\mathcal{G}'$, the oBN model under $\mathcal{G}^*$ is nested within the model under $\mathcal{G}'$, and both achieve the same superpopulation optimum: $Q_{\xi}(\theta_{\mathcal{G}^*}^* | \mathcal{G}^*) = Q_{\xi}(\theta_{\mathcal{G}'}^{\dagger} | \mathcal{G}')$. By the nested-model likelihood-ratio theory for pseudo-maximum likelihood (Rao and Scott, 1981; Lumley and Scott, 2014), the survey-weighted pseudo-log-likelihood ratio is

$$2 \left[\tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}') - \tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) \right] = O_p(1),$$

asymptotically distributed as a weighted mixture of chi-squared random variables with weights determined by the design effects of the added parameters. Meanwhile, $(K_{\mathcal{G}^*} - K_{\mathcal{G}'})\rho_n < 0$ with $|K_{\mathcal{G}^*} - K_{\mathcal{G}'}|\rho_n \rightarrow \infty$ since $\rho_n \rightarrow \infty$. Therefore

$$\widehat{\text{BIC}}(\mathcal{G}^*) - \widehat{\text{BIC}}(\mathcal{G}') = O_p(1) + (K_{\mathcal{G}^*} - K_{\mathcal{G}'})\rho_n \rightarrow -\infty$$

in probability.

2.3.2. Case 1B: $\mathcal{G}'$ has more edges than $\mathcal{G}^*$ but is not a supergraph. In this case, $\mathcal{G}'$ is misspecified: it both adds spurious edges and omits at least one true edge. The omission places $\mathcal{G}'$ in the regime of Case 2 below, where the likelihood gap dominates the penalty difference regardless of sign.

2.3.3. Case 2: $\mathcal{G}'$ is an underfit model (missing edges from $\mathcal{G}^*$). By Assumption 3 (Faithfulness), every edge in $\mathcal{G}^*$ corresponds to a genuine conditional dependence: $Q_{\xi}(\theta^* | \mathcal{G}^*) > Q_{\xi}(\theta^{\dagger} | \mathcal{G}')$. Let $\Delta = Q_{\xi}(\theta^* | \mathcal{G}^*) - Q_{\xi}(\theta^{\dagger} | \mathcal{G}') > 0$. By (5), the likelihood term is $N(\Delta + o_p(1))$, dominating the $O(\rho_n)$ penalty since $\rho_n/N \rightarrow 0$:

$$\widehat{\text{BIC}}(\mathcal{G}^*) - \widehat{\text{BIC}}(\mathcal{G}') = -2N(\Delta + o_p(1)) + O(\rho_n) \rightarrow -\infty.$$

2.3.4. Case 3: Same skeleton, not Markov-equivalent. This can only occur if $\mathcal{G}'$ and $\mathcal{G}^*$ differ in v-structures. By Assumption 3, $\mathcal{G}'$ implies at least one conditional independence not present in P_{ξ} , making it effectively misspecified (as in Case 2). □

2.4. *Proof of Theorem 2 (Orientation Consistency).* Let $\mathcal{G}^*$ be the true DAG and $\mathcal{G}'$ a Markov-equivalent DAG with $\mathcal{G}' \neq \mathcal{G}^*$. We show $\Pr(\ell(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) > \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}')) \rightarrow 1$.

2.4.1. *Step 1: Convergence of the pseudo-log-likelihood ratio.* Applying (5) to both $\mathcal{G}^*$ and $\mathcal{G}'$:

$$(7) \quad \frac{1}{N} \left[\tilde{\ell}(\hat{\theta}_{\mathcal{G}^*}^w | \mathcal{G}^*) - \tilde{\ell}(\hat{\theta}_{\mathcal{G}'}^w | \mathcal{G}') \right] \xrightarrow{p} Q_\xi(\theta_{\mathcal{G}^*}^* | \mathcal{G}^*) - Q_\xi(\theta_{\mathcal{G}'}^\dagger | \mathcal{G}'),$$

where $\theta_{\mathcal{G}^*}^* = \theta^*$ (since $\mathcal{G}^*$ is correctly specified) and $\theta_{\mathcal{G}'}^\dagger = \arg \max_\theta Q_\xi(\theta | \mathcal{G}')$ is the super-population KL-projection onto the model under $\mathcal{G}'$.

2.4.2. *Step 2: Identification of the limit as an expected KL divergence.* Expanding the definition of Q_ξ :

$$(8) \quad \begin{aligned} Q_\xi(\theta^* | \mathcal{G}^*) - Q_\xi(\theta_{\mathcal{G}'}^\dagger | \mathcal{G}') &= \mathbb{E}_\xi \left[\log P(\mathbf{X} | \mathcal{G}^*, \theta^*, \mathbf{Z}) - \log P(\mathbf{X} | \mathcal{G}', \theta_{\mathcal{G}'}^\dagger, \mathbf{Z}) \right] \\ &= \mathbb{E}_\mathbf{Z} \left\{ \mathbb{E}_\xi \left[\log \frac{P(\mathbf{X} | \mathcal{G}^*, \theta^*, \mathbf{Z})}{P(\mathbf{X} | \mathcal{G}', \theta_{\mathcal{G}'}^\dagger, \mathbf{Z})} \middle| \mathbf{Z} \right] \right\} \\ &\geq \mathbb{E}_\mathbf{Z} \left\{ \mathbb{E}_\xi \left[\log \frac{P(\mathbf{X} | \mathcal{G}^*, \theta^*, \mathbf{Z})}{P(\mathbf{X} | \mathcal{G}', \theta^\dagger(\mathbf{Z}), \mathbf{Z})} \middle| \mathbf{Z} \right] \right\} \\ &= \mathbb{E}_\mathbf{Z} \left[\text{KL} \left(P_\xi(\mathbf{X} | \mathcal{G}^*, \theta^*, \mathbf{Z}) \parallel P(\mathbf{X} | \mathcal{G}', \theta^\dagger(\mathbf{Z}), \mathbf{Z}) \right) \right], \end{aligned}$$

where the inequality in (8) holds because $\theta^\dagger(\mathbf{z})$ is the pointwise KL-minimizer given $\mathbf{Z} = \mathbf{z}$, so each inner expectation is at least as large as when evaluated at any other parameter. By Proposition 1, the final expected KL is strictly positive under the generic-parameter assumption on (θ^*, δ^*) : let $\Delta > 0$ denote its value.

2.4.3. *Step 3: Conclusion.* The pseudo-log-likelihood difference is $N[\Delta + o_p(1)] \rightarrow +\infty$ in probability, establishing the result. $\square$

REMARK 1. *No penalty term appears in this proof. The orientation comparison is purely a pseudo-likelihood ratio, and its consistency follows from the generic identifiability of the ordinal parameterization (Proposition 1) and the consistency of the pseudo-MLE (Proposition 2). This is the theoretical justification for Step 2 of the swa-oBn algorithm: removing the penalty from reversal moves does not compromise asymptotic correctness and eliminates the finite-sample sensitivity to heterogeneous ordinal scales.*

3. Polychoric Clustering Details. Figure 1 presents the weighted polychoric correlation matrix and hierarchical clustering dendrogram for the 13 CAHPS social need items. Items within each highlighted cluster 13 polychoric correlations $\rho \geq 0.75$.

4. Simulation Study 1: Edge Precision and Orientation Accuracy. Figure 2 presents edge precision (TP/(TP+FP)) for the swa-oBn versus naive oBN across design strength and signal strength conditions from Study 1 (Section 6.2 of the main text). Under mild and strong designs, the swa-oBn achieves substantially higher precision than the naive oBN (approximately 0.90–0.95 versus 0.70–0.75), indicating that the primary benefit of survey weighting is the elimination of spurious edges. Under no design, precision is identical for both methods.

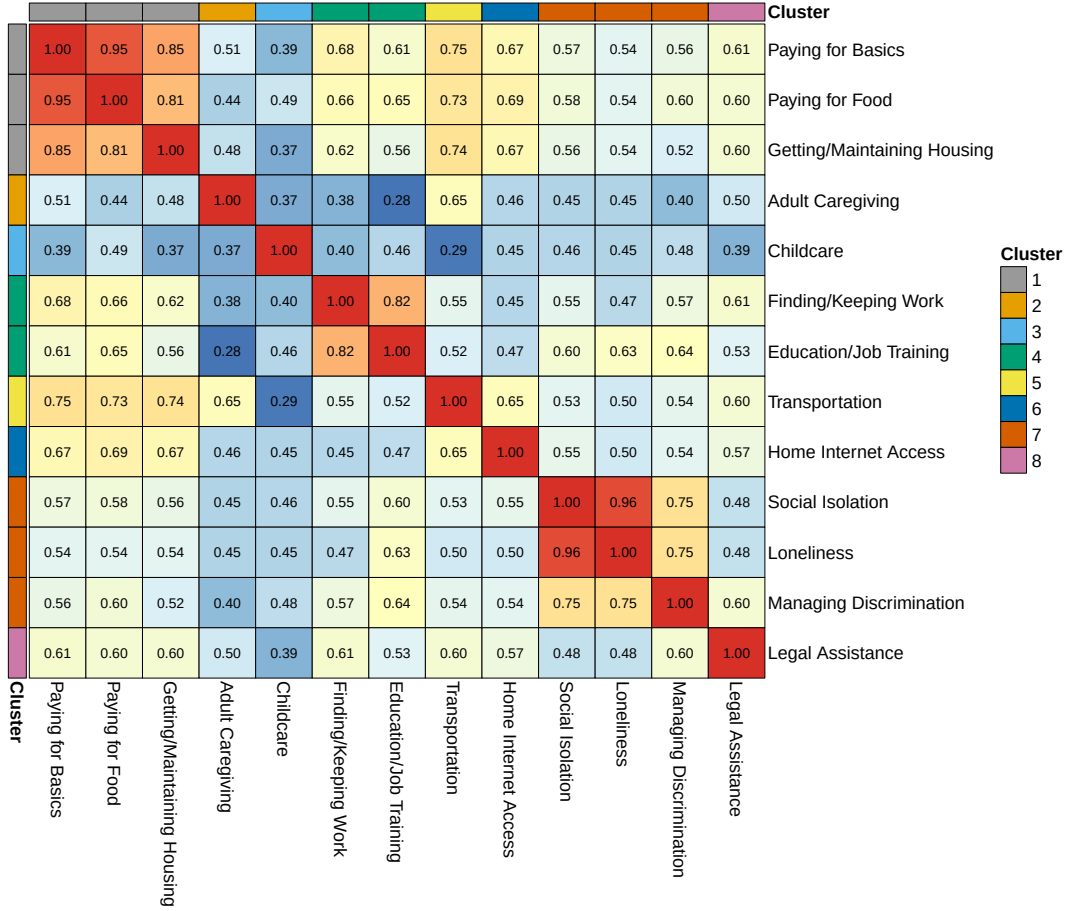


FIG 1. Polychoric correlation matrix and hierarchical clustering of 13 CAHPS social need items. Shaded blocks indicate clusters with $\rho \geq 0.75$: basic needs (food, housing, paying for basics), work/education, and discrimination/isolation/loneliness.

Figure 3 presents the correct orientation rate (fraction of true edges found and correctly oriented). Both methods achieve comparable orientation accuracy under no design; as the design strengthens, the naive oBN’s orientation rate degrades while the swa-oBN remains more stable. The pattern is consistent with the theoretical mechanism: survey weighting restores the distributional properties on which ordinal causal identification relies.

5. Simulation Study 3: Survey Weighting for Other Algorithm Families. To demonstrate that the distortion introduced by informative sampling is not specific to ordinal Bayesian networks, we evaluate survey-weighted and unweighted versions of three additional causal discovery algorithms: the PC algorithm (Spirtes, Glymour and Scheines, 2000), the FCI algorithm (Spirtes, Glymour and Scheines, 2000), and LiNGAM (Shimizu et al., 2006).

5.1. Setup. We generate a population of $N = 20,000$ units from a three-variable linear chain $X \rightarrow Y \rightarrow Z$ with continuous variables ($X, Z \sim N(0, 1)$, $Y = 0.8X + \varepsilon_Y$, $Z = 0.8Y + \varepsilon_Z$, $\varepsilon_Y, \varepsilon_Z \sim N(0, 1)$). From this population, Bernoulli samples are drawn under three design conditions (none, mild, strong), with oversampling based on the top quartile of the combined score of X and Z . Each scenario is replicated 100 times.

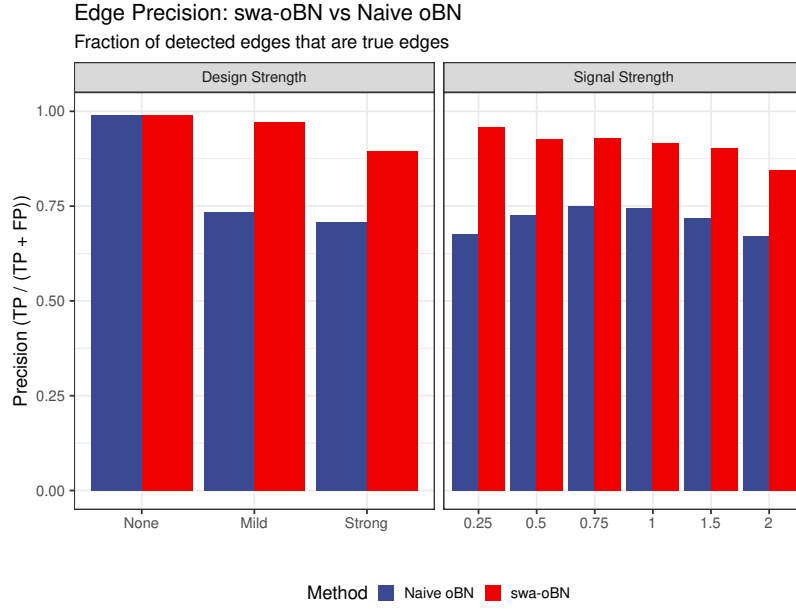


FIG 2. Edge precision ($TP/(TP+FP)$) of *swa-oBN* versus naive *oBN* across design strength (left) and signal strength (right) conditions. $q = 8$ nodes, 100 replications.

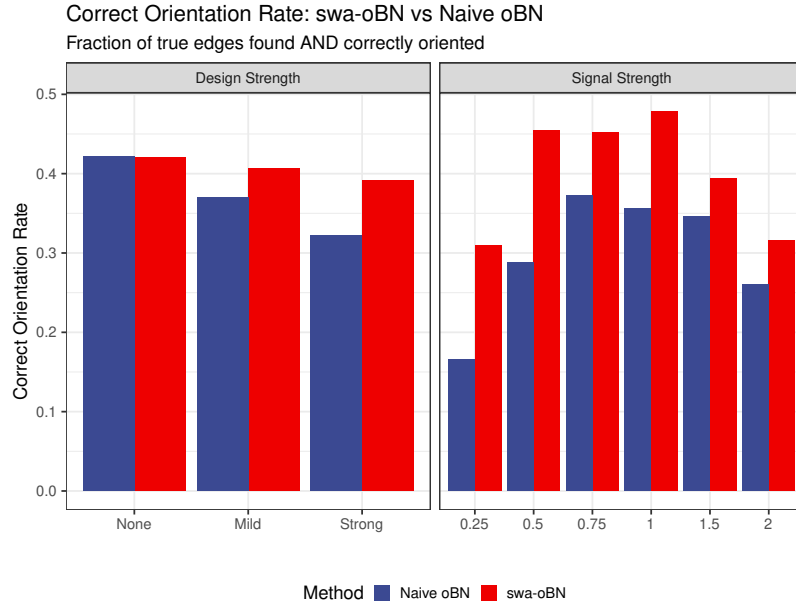


FIG 3. Correct orientation rate (fraction of true edges found and correctly oriented) across design strength (left) and signal strength (right) conditions. $q = 8$ nodes, 100 replications.

For the PC and FCI algorithms, survey weighting is implemented by replacing standard conditional independence tests with design-adjusted Wald tests using the `survey` package in R (Lumley, 2004). For LiNGAM, we implement a weighted pseudo-population resampling procedure that restores i.i.d. structure by resampling from the observed data proportional to survey weights prior to applying the standard ICA-based algorithm.

5.2. *Results.* Figure 4 presents the correct skeleton recovery rate across the three design conditions for all three algorithm families. Under simple random sampling (design = none), all six estimators (three algorithms $\times$ two weighting conditions) recover the correct skeleton in nearly all replications. Under mild and strong designs, the unweighted versions show a marked decline in skeleton recovery, while the survey-weighted versions maintain substantially higher correct recovery rates.

Figure 5 presents the spurious edge rate (proportion of replications in which a direct $X-Z$ edge is inferred). Under no design, spurious edges are rare for all methods. Under mild and strong designs, the unweighted versions infer the spurious $X-Z$ edge in a substantial fraction of replications — reflecting the sampling-induced marginal dependence between X and Z that persists even after conditioning on Y — while the survey-weighted versions maintain low spurious edge rates.

These results confirm that the survey-weighting problem documented in Studies 1 and 2 for the ordinal Bayesian network is a general phenomenon affecting all three major families of causal discovery algorithms.

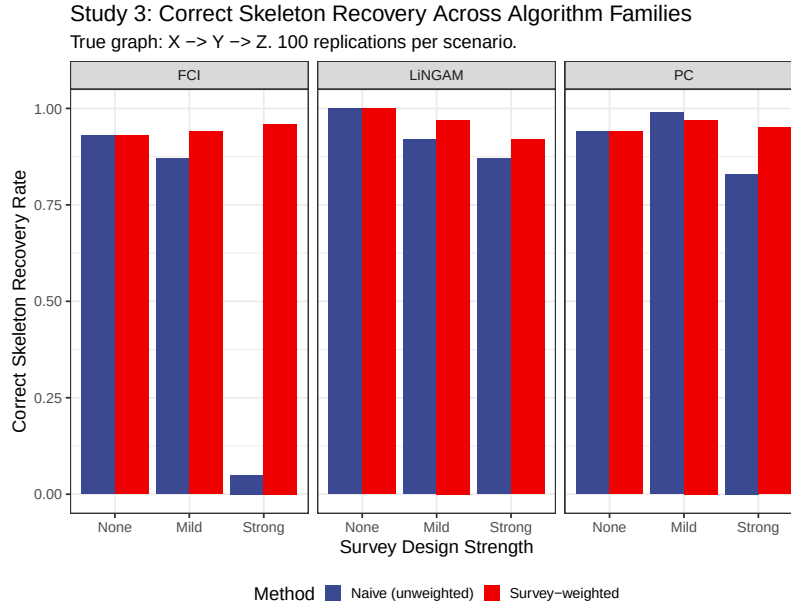


FIG 4. Study 3: Correct skeleton recovery rate for survey-weighted and unweighted versions of PC, FCI, and LiNGAM across three design conditions. True graph: $X \rightarrow Y \rightarrow Z$. 100 replications per scenario.

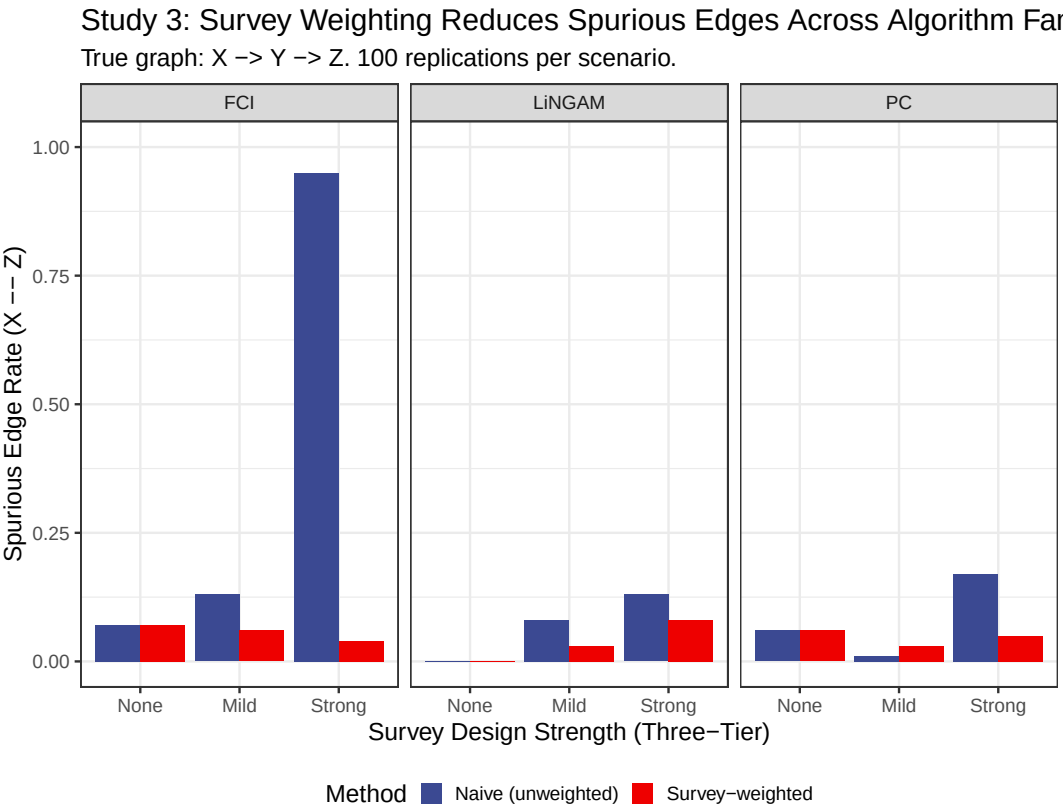


FIG 5. Study 3: Spurious X – Z edge rate for survey-weighted and unweighted versions of PC, FCI, and LiNGAM across three design conditions. True graph: $X \rightarrow Y \rightarrow Z$. 100 replications per scenario.

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6. Complete Bootstrap Edge Probability Matrix. Table 1 reports the full 10×10 matrix of bootstrap edge inclusion probabilities from the `swa-oBn` applied to the VHA CAHPS data, for both the weighted and the unweighted analyses of Section 7. Entry (j, k) gives $\hat{p}_{jk} = B^{-1} \sum_{b=1}^B \mathbf{1}(j \rightarrow k \in \hat{\mathcal{G}}^{(b)})$; rows index the parent (from) node and columns the child (to) node.

TABLE 1

Full 10×10 bootstrap edge inclusion probability matrix ($B = 250$ resamples). Each cell reports the weighted probability followed by the unweighted probability in brackets: $\hat{p}_{jk}^{wd} [\hat{p}_{jk}^{unwd}]$. Diagonal cells are omitted (“—”), as are the sixteen outcome-to-need cells forbidden by the blacklist, all of which have $\hat{p}_{jk} = 0$ in both specifications. Cells rendered in boldface indicate the ten edges whose weighted and unweighted inclusion probabilities fall on opposite sides of the 0.50 consensus threshold.

[illegible]

7. Weighted versus Unweighted Comparison. This appendix accompanies Section 7 of the main text, providing the side-by-side visualization and edge-level breakdown of the weighted versus unweighted $\text{swa-}\phi\text{Bn}$ consensus DAGs. Figure 6 presents the two graphs on a shared node layout so that differences in edge set and direction are directly comparable.

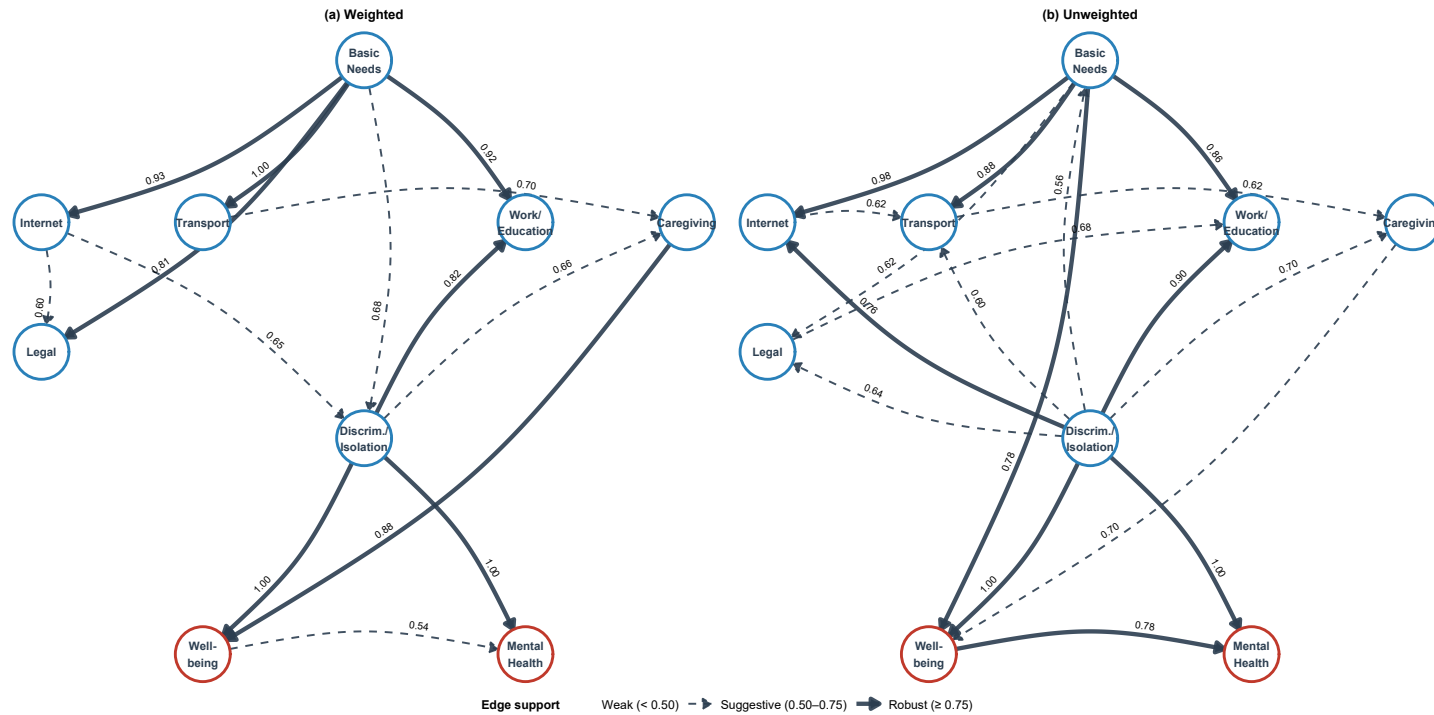


FIG 6. Consensus DAGs estimated with survey weighting (left) and without (right), on a shared node layout. Node positions, covariate adjustment, blacklist, restarts ($R = 10$), and bootstrap replicates ($B = 250$) are identical across panels; only the weights differ. Edge styling (dashed = suggestive, solid = robust) matches Figure ?? of the main text. Childcare is isolated in both specifications and is omitted for clarity.

Three specific differences sever the mediation pathway that the weighted analysis identifies as the paper’s principal substantive finding. First, the basic-needs–discrimination/isolation edge reverses direction: the weighted analysis selects basic needs $\rightarrow$ discrimination/isolation ($\hat{p} = 0.68$), while the unweighted analysis selects the reverse ($\hat{p} = 0.56$). Second, the unweighted analysis produces a direct basic-needs $\rightarrow$ well-being edge at $\hat{p} = 0.78$ that falls to 0.15 under weighting, spuriously bypassing the mediator. Third, internet $\rightarrow$ discrimination/isolation rises from $\hat{p} = 0.16$ unweighted to $\hat{p} = 0.65$ weighted, restoring a pathway the unweighted analysis misses. Together these changes produce the qualitatively different policy conclusion described in Sections 7 and 8.

8. Proportional Odds Diagnostics. The `swa-oBn` framework rests on the cumulative link specification of Assumption 4 of the main text, of which the proportional odds (PO) constraint is the most consequential working restriction: a single set of slope parameters is shared across the $L_j - 1$ cumulative thresholds for node j . This appendix reports diagnostic checks of PO for the application of Section 7 of the main text, documents the limitations of off-the-shelf diagnostics under a complex survey design, and presents a design-based sensitivity check for the regression whose orientation most directly drives our substantive conclusions.

8.1. Limitations of off-the-shelf diagnostics under a complex survey. The standard PO diagnostic is the [Brant \(1990\)](#) score test, implemented in R via `brant : : brant`. The test fits the cumulative link model via maximum likelihood and rejects PO if the between-threshold slope differences are jointly too large relative to the estimated covariance. In survey settings this procedure has two well-known failings. First, the estimator is a pseudo-MLE that treats observations as independently and identically distributed; the Hessian used to construct the score-test variance therefore omits the between-stratum and unequal-probability components of design variance and is anticonservative. Under design effect $deff$, the nominal test statistic is inflated by a factor of approximately $deff$, and at our sample size ($n_{\text{eff}} = 1,387$, $deff = 3.89$) this inflation can transform a modest PO departure into an apparently decisive rejection. Second, all B bootstrap replicates of the `swa-oBn` refit cumulative link regressions at parent sets that are themselves estimated; the nominal Brant distribution does not account for this parent-set stochasticity. We therefore present nominal Brant results as indicative rather than confirmatory, and follow them with a design-based analog for the regression in the discovered DAG whose orientation drives our principal finding.

8.2. Nominal Brant results. We report Brant tests under two parent specifications. *Pre-analysis* regresses each of the $q = 10$ ordinal nodes on the demographic covariates $\mathbf{Z}$ alone (age band and race-ethnicity-sex), a specification that makes no use of the discovered DAG. *Post-analysis* regresses each node on its estimated parent set from the consensus DAG (Figure 3 of the main text) together with $\mathbf{Z}$. At nominal $\alpha = 0.05$, the pre-analysis Brant test rejects PO in 1 of the 10 nodes; the post-analysis test rejects PO in 3 of 10. The increase in rejections from pre- to post-analysis is consistent with the Brant statistic inflating mechanically as the parent set grows (each added parent contributes $L_k - 1$ additional slopes to the joint test) and does not, on its own, indicate that the PO structure deteriorates conditional on the discovered parents. Given the anticonservatism documented in Section 8.1, we do not interpret these counts as evidence against the working model.

8.3. Working-model interpretation. Following [Lumley and Scott \(2015\)](#), we treat the PO cumulative link as an explicitly designated working model under complex-survey pseudo-likelihood. The generic identifiability result invoked throughout this paper (Proposition 1 of the main text) is a statement about the ordinal parameterization itself, not about whether the

data-generating mechanism exactly satisfies PO. Under the working-model view, the pseudo-MLE $\hat{\theta}^w$ consistently estimates the KL projection of the population distribution onto the PO model under the sampling design; the bootstrap inclusion probabilities $\hat{p}_{jk}$ describe the dependency structure of this projection, not of the unrestricted population. The substantive consequence is that the consensus DAG should be read as summarizing the dominant directed dependencies in the CAHPS data under the ordinal working model, which is the claim that the theory of Section 5 of the main text supports.

9. Survey Weight Distribution. Table 2 summarizes the distribution of survey weights in the CAHPS analytic sample. The design effect ($\text{DEFF} = n/n_{\text{eff}}$) quantifies the loss of effective sample size due to unequal weighting.

TABLE 2
Survey weight distribution and design effect.

	n	n_{eff}	DEFF	Min	Median	Max	CV
Overall	5,396	1,387	3.89	3.4	58.4	3,330.1	1.700

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